WESLEY JANES

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EDUCATION & HONORS

Mississippi State University, Bagley College of Engineering **Starkville, MS**
B.S. Aerospace Engineering, *Concentration in Astronautics* | **GPA: 3.5 / 4.0** *August 2021 – May 2026*
Shackouls Honors College | Cum Laude Graduate | Bulldog Recognition Award | Eagle Scout

ENGINEERING EXPERIENCE

Boeing **Wichita, KS**
Aerospace Structural Design Engineer – (777X Program) *June 2026 – Present*
• Designed primary fuselage structure for the 777X twin-aisle aircraft using CATIA, leveraging advanced part design and generative shape design workbenches to develop and modify production-ready models of aft lower lobe crown panels, side panels, skins, stringers, and frames while applying point design allowables, relational design, and Boeing structural design standards.
• Resolved complex fuselage manufacturing non-conformances through collaboration with shop-floor operations, liaison, manufacturing, and stress engineering, developing technically-sound design solutions and generating design change packages while maintaining structural integrity, producibility, configuration management, bill of material accuracy, and engineering traceability.
• Supported Boeing's design for six sigma product development process by contributing to design trade studies and delivering high-value engineering solutions that improve quality, reduce program risk, enhance manufacturing efficiency, and maximize long-term return on investment.

NASA STTR Partnership Research Initiative **Starkville, MS**
Undergraduate Researcher – (MSU A3RG Group x Embry-Riddle AU x WARD Aerospace) *August 2024 – August 2026*
• Engineered 16 advanced airfoil geometries using SolidWorks and resin-based additive manufacturing to investigate complex propeller-wing aerodynamic interactions, optimizing manufacturing while enabling rapid iteration of leading-edge modifications.
• Streamlined data collection by conducting subsonic wind-tunnel tests to characterize boundary layer and slipstream influence through Monte Carlo simulations with live LabVIEW data acquisition, ensuring high-fidelity aerodynamic profiling across varying Reynolds numbers and angles of attack.
• Developed custom MATLAB post-processing scripts to automate the calculation of lift, drag, and moment coefficients, establishing an empirical baseline to verify corresponding ANSYS CFD simulations.
• Co-authored a technical journal paper analyzing distributed electric propulsion configurations for publication in AIAA.

Xipiter Unmanned Aerial Systems Club **Starkville, MS**
Vice President / Design Team Lead *August 2023 – May 2026*
• Led a multidisciplinary design team of ~30 student-engineers in the structural development of a custom UAS using SolidWorks, designing airframe structures, avionics mounting provisions, and small component interfaces to meet structural and packaging requirements.
• Directed airframe manufacturing and assembly of composite, sheet-metal, and 3D-printed components, developing engineering drawings and CAD models with GD&T, while applying design for manufacturability principles, and optimizing slicing to ensure structural integrity of the final airframe.
• Coordinated weekly cross-functional design reviews between CAD, avionics, software, and manufacturing teams to resolve interface conflicts, incorporate subsystem requirements, and mature the airframe from preliminary design through release.

Aerostructure Design of Stiffened Panels for Multiple Failure Modes **Starkville, MS**
Design Lead *August 2025 – December 2025*
• Engineered c- and hat-stiffened fuselage panel configurations representative of large aircraft structures using SolidWorks and Abaqus FEA, comparing stress, buckling, and failure behavior to evaluate structural efficiency for future fuselage designs.
• Optimized manufacturing and assembly processes by applying GD&T and design for manufacturability principles to sheet-metal cutting, bending, drilling, and riveting of structural test articles.
• Validated structural performance by correlating UTM compression testing with Abaqus FEA and analytical predictions, evaluating failure behavior and load-carrying capability across competing stiffener configurations.

Systems & Mission Design Analysis of the X-15 Hypersonic Vehicle **Starkville, MS**
Modeling & Simulation Lead *August 2025 – December 2025*
• Translated historical mission requirements into technical drawings to develop a detailed SolidWorks airframe model of the X-15 airframe and primary structural geometry.
• Performed 3D thermal analysis using ANSYS Fluent at hypersonic flight conditions up to Mach 8, evaluating aerodynamic heating and temperature distributions across the fuselage, nose, and wing leading edges to assess thermal ablative coating protection requirements.
• Presented technical findings to nationally recognized museums, correlating modern computational analysis with historical NASA flight data and design requirements to validate vehicle performance.

TECHNICAL SKILLS

Software & Testing: CATIA / 3DEXPERIENCE, SolidWorks, AutoCAD, Abaqus, ANSYS Mechanical / Fluent, MATLAB, Python, C/C++, Git, LabVIEW, STK, Linux, Microsoft & Adobe Suite, Wind Tunnel Testing, Data Acquisition.
Manufacturing & Engineering: Additive / Subtractive Manufacturing, GD&T / HMBD, Engineering Drawings, Design Reviews, Design for Six Sigma, Design for Manufacturability, CNC Machining, Composites & Sheet Metal Fabrication.